Grover's Search and Quantum Counting

Project 6 - PHYS-541 Quantum Computing

Presented by: Victor Peucelle

École Polytechnique Fédérale de Lausanne
Academic Year: 2025–2026

Professor: Vincenzo Savona
Assistants: Sara Alves dos Santos, David Linteau, Shao Chiew

December 31, 2025

Project Overview & Objectives

This project focuses on the implementation and analysis of Grover's Search and Quantum Counting algorithms.

1. **Theory:** Detailed explanation of Grover's algorithm and Quantum Counting (based on Nielsen & Chuang).
2. **Complexity:** Analysis of quantum speedup limits and NP complexity class.
3. **Implementation:** Construction of Oracles (with and without Ancilla) on IBM Qiskit.
4. **Noise Analysis:** Performance study under simulated noise and circuit depth limitations.

1. Grover's Algorithm: The Mechanism

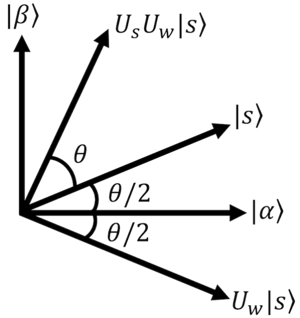
Goal: Find marked element(s) x such that $f(x) = 1$ in a database of size $N = 2^n$.

The algorithm iterates the **Grover Operator** $G = (2|\psi\rangle\langle\psi| - I)O$:

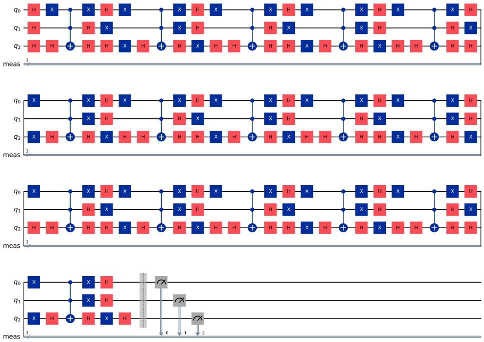
- **Oracle (O):** Flips the phase of the solution $|x\rangle$.
- **Diffuser:** Inversion about the mean.

Geometric Interpretation: Rotates the state vector by angle θ towards the target state $|\beta\rangle$.

$$\theta \approx 2\sqrt{m/N}$$



Circuit Implementation ($n = 3$)



1. Initialization

Apply $H^{\otimes n}$ to create equal superposition of all $N = 8$ states.

2. Oracle (U_w)

Flips the phase of the target state (e.g., $|101\rangle$).

3. Diffuser (U_s)

Inversion about the mean. Increases amplitude of the marked state.

Theoretical Performance

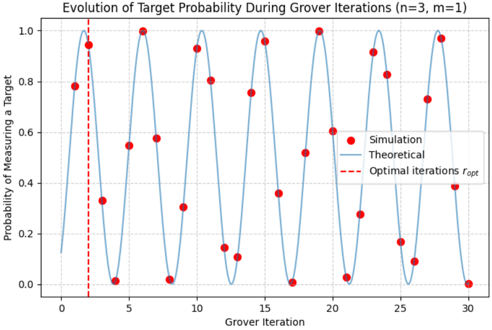
Number of Iterations: To maximize the probability of measuring a marked state, perform:

$$r \approx \frac{\pi}{4}\sqrt{\frac{N}{m}}$$

Probability of Success After r Iterations: $P_{\text{success}} = \sin^2\left((2r+1)\frac{\theta}{2}\right)$

Example: $n = 3$ ($N = 8$), $m = 1$:

- $\theta \approx 0.707$ radians.
- Optimal iterations: $r \approx 1$.
- $P_{\text{success}} \approx 1$.



1. Quantum Counting Algorithm

Problem: Grover's optimal iterations r depend on m. What if m is unknown?

The Solution

Combine Grover operator G with **Quantum Phase Estimation (QPE)**.

- The eigenvalues of G are $e^{\pm i\theta}$.
- QPE estimates the phase θ.
- Since $\sin(\theta/2) = \sqrt{m/N}$, we can deduce m:

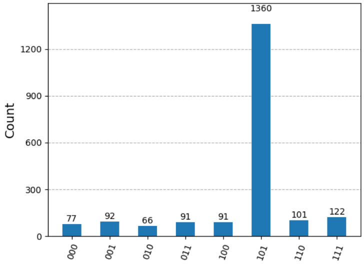
$$m \approx N \sin^2(\theta/2)$$

Quantum Counting on Real Hardware (IBM Turing)

Experimental Setup

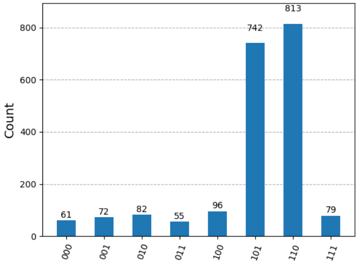
- **Backend:** `ibm_turing` (127-qubit Eagle processor).
- **Parameters:** $n = 3$ (Search Space), 2000 Shots.
- **Objective:** Estimate m (number of solutions) via Phase Estimation.

Experiment A: 1 Solution



Measured Phase Distribution ($m = 1$)

Experiment B: 2 Solutions



Measured Phase Distribution ($m = 2$)

2. Grover and Computational Complexity

Quadratic Speedup vs Exponential Speedup

- Classical Search: $\mathcal{O}(N)$
- Grover Search: $\mathcal{O}(\sqrt{N})$

Can Grover solve NP-Complete problems efficiently?

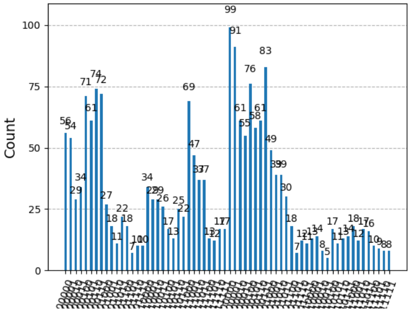
Limitation

NP-Complete problems (like SAT) have search spaces of size $N = 2^n$.

- Brute force: 2^n operations.
- Grover: $\sqrt{2^n} = 2^{n/2}$ operations.

While $2^{n/2}$ is a significant improvement, it is **still exponential**. Grover does not place NP-complete problems into the BQP class.

Failure of QPE on Real Hardware



Observation: While Grover's Search ($\mathcal{O}(\sqrt{N})$ depth) yielded results on IBM Heron, Quantum Counting produced uniform noise.

Why?

- QPE requires applying Controlled-G^{2^k}.
- Controlling the Grover operator drastically increases the CNOT count.
- Total depth exceeds the qubit coherence time (T_2).

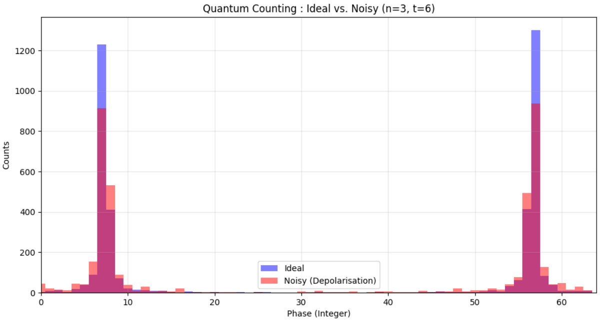
Conclusion

Gap between 'Search' (NISQ-friendly) and 'Estimation' (Requires Error Correction).

Quantum Counting Results (Simulation)

Setup:

- Search space: $n = 3$ qubits ($N = 8$).
- Precision: $t = 6$ counting qubits (allows distinguishing phases with accuracy 2^{-6}).
- Noise Model: Depolarizing error ($\epsilon = 10^{-5}$).



Observation: The dominant peak corresponds to the eigenvalue θ. Using the formula $m \approx N \sin^2(\theta/2)$, the algorithm correctly identifies the number of solutions m, even with simulated noise.

3. Oracle Implementation Strategy

Task: Implement $f(x) = 1$ for $m = 1$ or $m = 2$ targets.
We explored two approaches for the Multi-Controlled-X (MCX) gates:

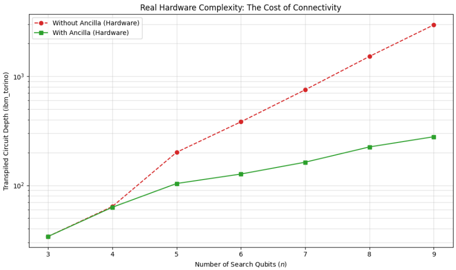
A. With Ancilla Qubits

- Uses extra qubits to decompose Toffoli gates.
- **Pros:** Linear depth scaling $\mathcal{O}(n)$.
- **Cons:** Requires more qubits.

B. Without Ancilla

- Uses complex decompositions of MCX gates.
- **Pros:** Saves qubits.
- **Cons:** Depth grows exponentially, creating massive noise.

3. Hardware Complexity: The Cost of Connectivity



Experimental Setup:

- **Backend:** `ibm_torino` (133-qubit Heron).
- **Metric:** Transpiled Depth (Log Scale).
- **Optimization Level:** 3 (Max).

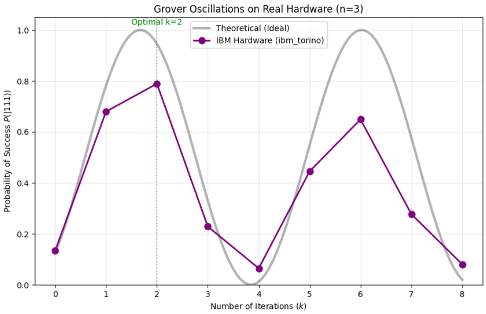
Analysis:

1. **No Ancilla (Red):** Uses `MCXGrayCode`. Depth grows exponentially.
2. **With Ancilla (Green):** Uses `MCXVChain`. Depth grows linearly.

Key Finding

Without ancillas, the transpiler must insert massive amounts of **SWAP gates** to satisfy the limited connectivity of the chip, making the circuit useless for $n > 5$.

4. Hardware Performance: Grover Oscillations



Experimental Protocol:

- **Target:** $|111\rangle$ ($n = 3$).
- **Shots:** 2000 per iteration step.
- **Transpilation:** Optimization Level 3 (Aggressive gate reduction).

Key Observations:

1. **Validation:** The hardware (purple) closely follows the theoretical sinusoidal curve (black).
2. **Optimal Peak:** Maximum probability occurs at $k = 2$, matching the theory exactly.
3. **Damping:** The peak < 1.0 and decays over time. As k increases, circuit depth increases, accumulating gate errors and decoherence.

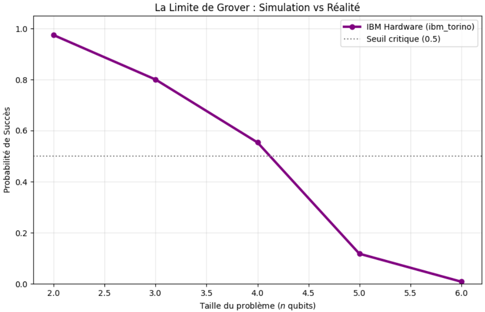
4. Performance Under Noise

The Challenge: Small $\theta/2 \approx \sqrt{m/N}$ implies large number of iterations r.

$$\text{Circuit Depth} \propto r \times \text{Oracle Depth}$$

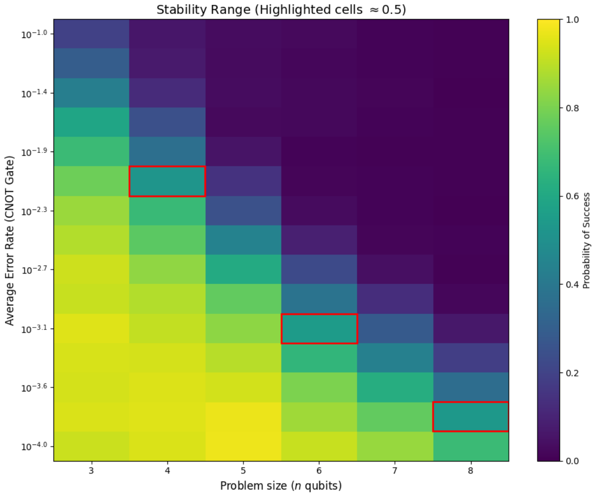
Simulation Results:

- For small n ($n \leq 4$): High fidelity.
- For larger n: Success probability drops below classical threshold.
- **Limit:** On current NISQ models, coherence time limits usage to small instances.



The "Tipping Point" (Red Boxes):

- Highlighted cells indicate $P \approx 0.5$.
- This is the **critical threshold**. Beyond this line (top-right), noise dominates the quantum signal.
- **Observation:** For current hardware error rates ($\sim 10^{-2}$), we are limited to very small n.



Conclusion

- **Verification:** We successfully verified Grover's quadratic speedup and the ability to estimate m via Quantum Counting.
- **Implementation Trade-offs:** Using Ancilla qubits is crucial on real hardware to keep circuit depth manageable.
- **Hardware Limits:** Noise analysis shows that Grover's algorithm requires deep circuits, making it challenging for current NISQ devices beyond small n.

Thank you for your attention.